Language Technology

Chapter 1: An Overview of Language Processing

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Applications of Language Processing

A brief history:

- Spelling and grammar checkers: *MS Word*, email clients, etc.
- Text indexing and information retrieval on the Internet: *Google*, *Microsoft Bing*, *Yahoo*, and software such as *Apache Lucene*
- Machine translation: *Google Translate*, *DeepL*, *Bing Translator*, etc.
- Spoken interaction: Apple Siri, Google Assistant, Amazon Echo
- Speech dictation of letters or reports: *Windows*, *macOS*

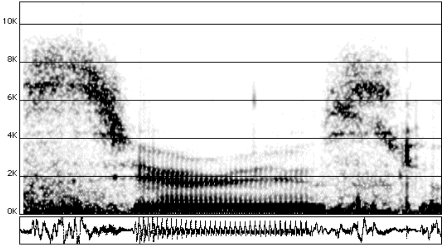
Applications of Language Processing (continued)

- Direct translation from spoken English to spoken Swedish in a restricted domain: *SRI* and *SICS*
- Voice control of domestic devices such as tape recorders (*Philips*) or disc changers (*MS Persona*)
- Conversational agents capable of dialogue and planning: *TRAINS*
- Spoken navigation in virtual worlds: *Ulysse*, *Higgins*
- Generation of 3D scenes from text: *Carsim*
- Question answering: *IBM Watson* on *Jeopardy!*

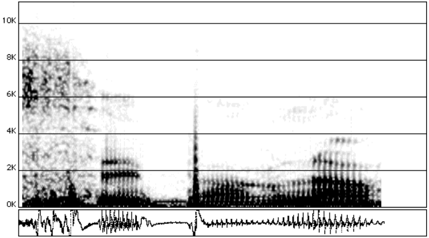
Linguistic Objects and Vocabulary

- Sounds
- Phonemes
- Words and morphology
- Syntax and grammatical functions
- Semantics
- Dialogue

Sounds and Phonemes



Serious



C'est par là ('It is that way')

Lexicon and Parts of Speech

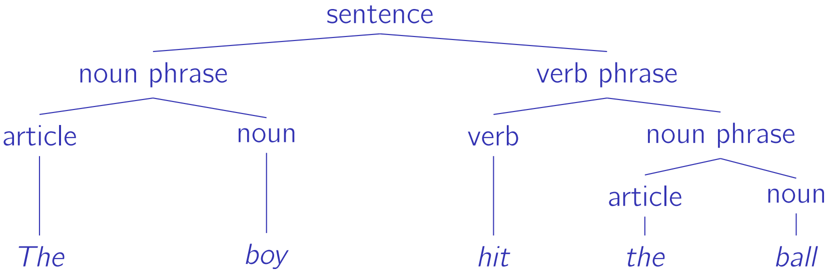
The big cat ate the gray mouse

The/article *big*/adjective *cat*/noun *ate*/verb *the*/article *gray*/adjective *mouse*/noun
Le/article *gros*/adjectif *chat*/nom *mange*/verbe *la*/article *souris*/nom *grise*/adjectif
Die/Artikel *große*/Adjektiv *Katze*/Substantiv *ißt*/Verb *die*/Artikel *graue*/Adjektiv *Maus*/Substantiv

Morphology

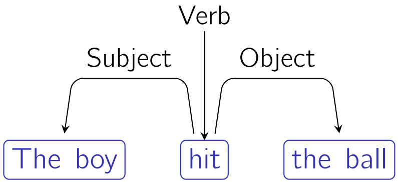
Word	Morphemes	Root form and grammatical features
worked	*work* + *ed*	*work* + verb + past tense
travaillé	*travaill* + *é*	*travailler* + verb + past participle
gearbeitet	*ge* + *arbeit* + *et*	*arbeiten* + verb + past participle

Syntactic Tree



Syntax: A Classical View

A dependency graph with grammatical functions



Semantics

In contrast to syntax:

1. Colorless green ideas sleep furiously.
2. *Furiously sleep ideas green colorless.

Determining the logical form:

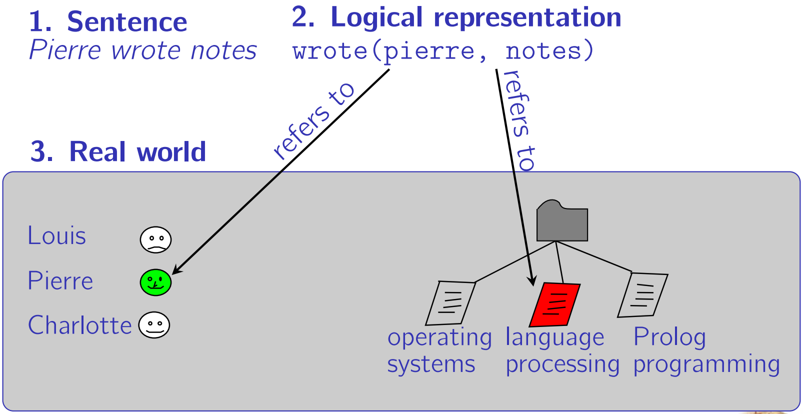
Sentence	Logical representation
Frank is writing notes	`writing(Frank, notes).`
François écrit des notes	`écrit(François, notes).`
Franz schreibt Notizen	`schreibt(Franz, Notizen).`

Lexical Semantics

Word senses:

1. **note** (*noun*): a short piece of writing
2. **note** (*noun*): a single sound at a particular pitch
3. **note** (*noun*): a piece of paper money
4. **note** (*verb*): to take notice of
5. **note** (*noun*): of note — of importance

Reference


1. Sentence
Pierre wrote notes
2. Logical representation
wrote(pierre, notes)
refers to
refers to
3. Real world
Louis
Pierre
Charlotte
operating systems
language processing
Prolog programming

Ambiguity

Many analyses are ambiguous, which makes language processing difficult. Ambiguity occurs at every level: speech recognition, part-of-speech tagging, parsing, etc.

Example of an ambiguous phonetic transcription:
The boys eat the sandwiches
which may also correspond to:
The boy seat the sandwiches; *the boy seat this and which is*; *the buoys eat the sand which is*

Models and Tools

- Linguistics has produced an impressive set of theories and models.
- Early theories were often inadequate, and data (corpora, dictionaries, annotated resources) were scarce.
- Models and tools have matured, and large datasets are now widely available.
- The main tools include regular expressions, logic, statistics, and machine learning.
- Language processing generally requires significant computational power and modern GPU architectures.
- These advances have led to massive improvements across most areas of natural language processing (NLP).

An Application Example: IBM Watson

- IBM Watson: a landmark system that answered questions better than any human competitor
- Video: https://www.youtube.com/watch?v=WFR3lOm_xhE

- IBM Watson relied on knowledge extraction from large text collections (Wikipedia, New York Times archives, etc.) and multiple language processing modules.

Two Viewpoints

Given a task to solve, natural language processing has followed two main approaches:

1. Try to understand human language processing and reproduce it
   - For example, model all the syntactic and semantic phenomena involved in translating a sentence
2. Design a system that replicates human performance on existing examples
   - For example, produce translations that match those of human translators, regardless of the internal process

The second viewpoint is now largely dominant, as validated by benchmarks that are mostly agnostic to the underlying methods.

Datasets

Digitization has been a key driver of progress in NLP.
Huge datasets are now available to researchers and companies.
We can divide them into two categories:

- Annotated text: for example, IMDB movie reviews labeled as positive or negative for sentiment classification (`https://ai.stanford.edu/~amaas/data/sentiment/`)
- Raw text corpora: Wikipedia, book archives, Common Crawl (`https://commoncrawl.org/`), and the Colossal Clean Crawled Corpus (C4, 156 billion English tokens) (`https://huggingface.co/datasets/allenai/c4`)

Training

Given a parametrizable mathematical model, datasets are used to adjust its parameters:

1. Annotated corpora → supervised training
2. Raw corpora → unsupervised or self-supervised training, for example to predict:
    1. the next word in a sequence (autoregressive language model), or
    2. missing words in a sentence (masked language model)

Large language models (LLMs) are trained on web-scale corpora.

Corpora as Knowledge Sources

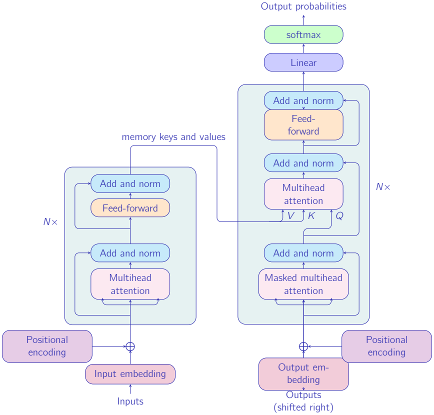
Corpora (text collections) are repositories of both knowledge and linguistic regularities:

1. If large enough, a raw corpus can capture the semantics of words and texts.
2. Fitting a model on a raw corpus is often not sufficient for a final application. The resulting model can then be further adapted to a downstream task.

This two-stage process is known as **pretraining** followed by **fine-tuning**.

The Transformer

The transformer architecture currently defines the state of the art for many NLP tasks.

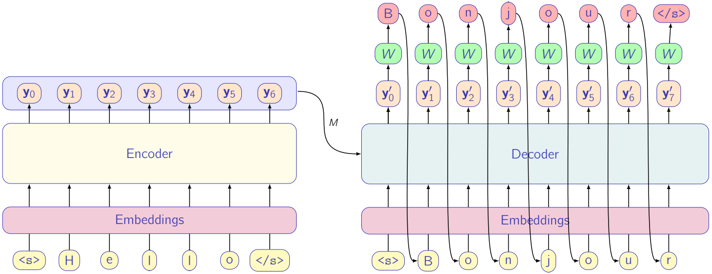


From Nugues, *Python for Natural Language Processing*, https://link.springer.com/chapter/10.1007/978-3-031-57549-5_17

Translation

Hello → *Bonjour*

Using an encoder–decoder architecture: an autoregressive process



From Nugues, *Python for Natural Language Processing*, `https://link.springer.com/chapter/10.1007/978-3-031-57549-5_17`

Stanford Question Answering Dataset (SQuAD)

- Consists of 100,000 questions paired with Wikipedia paragraphs that contain the answers
- The answer is a text span (factoid QA)
- Extended by SQuAD 2.0 with unanswerable questions
- SQuAD sparked intense competition — see the leaderboard: `https://rajpurkar.github.io/SQuAD-explorer/`

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?
graupel

Where do water droplets collide with ice crystals to form precipitation?
within a cloud

Figure 1: Question-answer pairs for a sample passage in the SQuAD dataset. Each of the answers is a segment of text from the passage.

From Rajpurkar et al., *SQuAD: 100,000+ Questions for Machine Comprehension of Text*, 2016

Architecture Evolution

Question answering systems evolved from pipelines of linguistic modules to large language models.

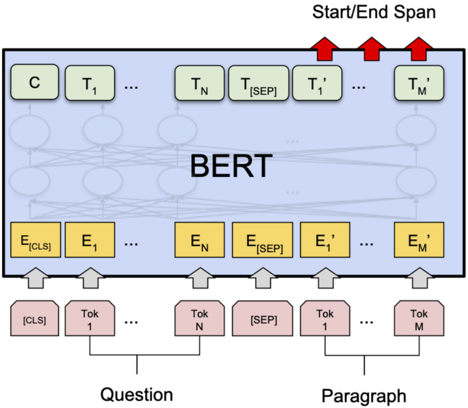
Simplified architecture of IBM Watson:

Question parsing and classification:
Syntactic parsing, entity recognition, answer type classification

Document retrieval and passage ranking:
Indexing, vector space model

Answer extraction and ranking:
Answer parsing, entity recognition

Question Answering with a Transformer



(c) Question Answering Tasks:
SQuAD v1.1

From Devlin et al., *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*, 2019

GLUE Evaluation

To evaluate claims rigorously, the research community relies on benchmarks such as SQuAD.

The General Language Understanding Evaluation (GLUE) benchmark (`https://gluebenchmark.com/`) covers three main types of tasks:

1. Sentence or text classification
2. Classification of relations between sentence pairs (similarity, entailment, etc.)
3. Question answering

Evaluation of Large Language Models

Machine translation evaluation: https://www2.statmt.org/wmt26/

Large language models are evaluated on many benchmarks, for example:

- Massive Multitask Language Understanding (MMLU)
  https://huggingface.co/datasets/cais/mmlu
- Grade School Math 8K (GSM8K)
  https://huggingface.co/datasets/openai/gsm8k
- HumanEval for coding
  https://huggingface.co/datasets/openai/openai_humaneval

LiveBench aggregates several benchmarks with regularly updated data:
https://livebench.ai/
See also https://artificialanalysis.ai/

A Word on Computing Resources

Progress in NLP is not free: large models now contain tens or hundreds of billions of parameters (e.g., Meta's Llama models).

From Strubell et al. (2019)

Consumption	CO_2e (lbs)
Air travel, 1 person, NY↔SF	1984
Human life, avg, 1 year	11,023
American life, avg, 1 year	36,156
Car, avg incl. fuel, 1 lifetime	126,000
Training one model (GPU)	
NLP pipeline (parsing, SRL)	39
w/ tuning & experiments	78,468
Transformer (big)	192
w/ neural arch. search	626,155

Table 1: Estimated CO_2 emissions from training common NLP models, compared to familiar consumption.[1]

From Touvron et al. (2023) on training Llama

Finally, we estimate that we used 2048 A100-80GB for a period of approximately 5 months to develop our models. This means that developing these models would have cost around 2,638 MWh under our assumptions, and a total emission of 1,015 tCO_2eq. We hope that releasing these models will help to reduce future carbon emission since the training is already done, and some of the models are relatively small and can be run on a single GPU.